

Personal Statement

Applicant: Nauval Zulfikar

Programme: PhD in Computer Science & AI, Tel Aviv University

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Growing up in Bandung, Indonesia, I was always fascinated by the intricate dance of logistics and transportation that unfolded daily in our bustling city. The streets were alive with a constant flow of vehicles, each journey a small part of a larger, complex system. This fascination only deepened during my undergraduate studies in Japan, where I observed the seamless integration of technology and human behaviour in transportation systems. It was during these formative years that I realised the profound impact that well-designed systems could have on efficiency and quality of life. This realisation has driven my academic and professional pursuits, culminating in my decision to apply for a PhD at Tel Aviv University.

Academic Foundations

My academic journey began at Ritsumeikan Asia Pacific University in Japan, where I earned a Bachelor of Business Administration in Marketing. Here, I developed a strong foundation in consumer behaviour, marketing research, and business data analysis, supported by the JASSO Scholarship and a 50% Tuition Reduction Scholarship. These disciplines provided me with a unique perspective on the behavioural aspects of logistics and transportation systems.

I further honed my analytical skills at Aston University, where I pursued an MSc in Business Analytics, supported by the Aston Enterprise Scholarship 2023. My coursework, which included Decision Models, Software Analytics, and Effective Management Consultancy, equipped me with a robust quantitative toolkit. My dissertation, supervised by Dr. Ammar Al-Bazi, focused on enhancing supply chain information sharing systems through blockchain and DeBERTa-v3 transformer NLP, earning a First Class distinction.

Technical and Research Experience

During my MSc, I conducted a dissertation titled "Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach." This project involved fine-tuning the DeBERTa-v3 transformer on multi-tier supplier review data and developing an Ethereum smart-contract architecture to improve transparency in supply chains. This work not only deepened my understanding of NLP and blockchain technologies but also highlighted the potential of LLMs in transforming supply chain management.

In addition to my dissertation, I have developed a portfolio of applied LLM projects. Notably, I created the LLM-Generated Adaptive Shipper Decision Rules system, which leverages OpenAI's gpt-4o-mini and retrieval-augmented generation techniques to produce literature-grounded decision rules for shippers. This project, along with others like the CV-JD Suitability Checker and the Crowd-Shipping ABM, showcases my ability to apply advanced machine learning techniques to real-world problems.

Industry Experience

My role as a Data Scientist and Project Manager at the Public Works & Spatial Planning Department in Bandung has been instrumental in shaping my understanding of the practical applications of my research. Here, I automated ETL pipelines and developed self-service dashboards, reducing project delivery times by 45% and improving data efficiency by 20%. This experience taught me the importance of data governance and privacy, as well as the challenges of implementing machine learning solutions in a government setting.

At Bank Muamalat Indonesia, I led the development of NLP models for credit portfolio analysis and risk scoring, which sparked my interest in the post-deployment behaviour of models and the technical preconditions of Responsible AI. These experiences have provided me with a solid foundation in applied machine learning and have informed my research interests in NLP and AI.

Why Tel Aviv University

Tel Aviv University is renowned for its cutting-edge research in natural language processing and AI, areas that align perfectly with my academic and professional interests. I am particularly drawn to the work of Professor Jonathan Berant, whose research on question answering and reasoning in NLP inspires me. His recent paper, [TODO: verify on <https://scholar.google.com/citations?q=Jonathan+Berant>], exemplifies the innovative approaches I aspire to contribute to. The opportunity to work under his guidance would be invaluable in advancing my research skills and contributing to the field.

What I Bring and What I'm Building

I bring a strong foundation in business analytics, NLP, and machine learning, supported by practical experience in both academic and industry settings. My technical skills include proficiency in Python, PyTorch, and HuggingFace Transformers, as well as experience with DeBERTa-v3 fine-tuning and LLM applications. However, I am still building expertise in large-scale pretraining and formal theoretical ML depth, areas I am eager to develop further during my PhD.

Career Vision

My ultimate goal is to become a leading researcher in the field of AI, with a focus on developing innovative NLP solutions that address real-world challenges. I envision a career where I can bridge the

gap between academia and industry, leveraging cutting-edge research to drive practical advancements in technology. Pursuing a PhD at Tel Aviv University is a crucial step towards achieving this vision, providing me with the opportunity to collaborate with world-class researchers and contribute to the advancement of AI and NLP.